Project

Due: Part 1: Tuesday, 04/12/26 by 11:59 PM
Part 2: Tuesday, 04/28/26 by 11:59 PM

Score: Part 1 /10 Part 2 /10 Total /20

Project Part 1 Directions

Deliverables

- Push code (including `.Rmd` file), data sets, and report to your team's GitHub repository by 11:59 PM on **04/12**.
- Every team member must upload a copy of the report that has all team members' names to Canvas by 11:59 PM on **04/28**.
- Be sure to include the URL to your team's repository on the first page after the title and author's names. Your team's repository is *not* a reference.

Grading

- You will be graded on:
  1. Your ability to clearly articulate the question that you intend to answer.
  2. How thorough your exploratory analysis is.
  3. Readability of your `R` code, including comments, and report.
- The report should not exceed 7 pages, excluding references; otherwise you will be penalized. However, there's no minimum page length (the shorter the better ☺).
- There are no bonus point opportunities.
- I will deduct 4 points for every hour (rounded up) that your deliverables are late, either to GitHub or Canvas.

Requirements

- You can complete the project individually or in groups of up to 3.
- Pick *up to* 3 preliminary questions that can be answered using data:
  1. The question(s) can explore any type of parameter – center, spread, proportion, etc. — but needs to focus on one or two specific parameters that can be analyzed with a single, stand-alone statistical test.
  2. Part 2 of the project will NOT include linear regression, logistic regression, time series, or any other predictive modeling method.

- You must use real data:
  1. Data should come from one of the following online sources:
     (a) Government agencies
     (b) Reputable global organizations (WHO, UN, World Bank, etc.)
     (c) Reputable news organizations (including FiveThirtyEight)
     (d) Reputable non-profit organizations
     (e) Reputable academic researchers (e.g. your PI if you support academic research) and/or academic journals
     (f) Sports leagues
  2. Include the data set in your repo and URL to the source of your data set in the References section of your report.
  3. Note that GitHub has storage limits, so if your data set is too large to upload in compressed format, either subset the data or use a different data set.
  4. If you have trouble finding data, your topic may be too ambitious for a 5-week project.
- Part of this project is getting acquainted with the technology.
  1. You can (and frankly, should) use generative AI to assist you.
  2. You have more sophisticated tools at your disposal, so I will expect more polished deliverables. Also, I will *not* help you debug!
  3. If I can't reproduce your results using the files in your repo (as of 11:59 PM on 04/12), then I will deduct points. This is where students typically lose the most points.

Part 1 Report

- The report must be a `PDF` file knit directly from the provided project template `.Rmd` file. The report should include the following labeled sections:
  1. Introduction: Introduce readers to the chosen topic and research question.
  2. Data Summary:
     (a) If the data represent a population, explain how the data were collected.
     (b) If the data represent a sample, explain how the data were selected and collected.
     (c) Explain any data modifications after collection and the reasons behind these changes.
     (d) Discuss potential issues with the data and their possible impact.
     (e) Explain why the data are appropriate to answer the research question.
  3. Exploratory Analysis:
     (a) Include at least two numerical and two graphical summaries.
     (b) Display all of the chosen summaries in this section and within the 7-page report limit.
     (c) Appropriately label and title all elements of all graphical summaries.
     (d) Numerical summaries can be displayed graphically by displaying the numeric values on a corresponding graphic. Two summaries combined in this way are considered two of the four summaries.

(e) Create all summaries in `R` and all graphical summaries using `ggplot2`.
(f) Do NOT perform any statistical inference.

4. Conclusions:
   (a) Interpret the chosen summaries and make initial conclusions based on the information displayed in them.
   (b) Interpret the values, trends, and patterns displayed in the chosen summaries statistically.
   (c) Explain what the values, trends, and patterns displayed in the chosen summaries may mean or imply in the context of the question.
   (d) Summarize the conclusions in the context of the topic of interest.
5. References:
   (a) This section is not considered part of the 7-page report length limit.
   (b) Begin this section on a new page.
   (c) Include sources used for background research and the data sources.
   (d) If you used AI to produce any portion of any file in your repository, you must provide attribution in accordance with UVA guidelines.

Advice

- Pick a topic that you are somewhat knowledgeable about. For example, I don't know anything about hockey, so I wouldn't analyze hockey player performance.
- Do not bite off more than you can chew. Properly scope the question to be able to answer it in 4 weeks.
- Budget extra time for writing the actual report.

Project Part 2 Directions

Deliverables

- Push code (including `.Rmd` file), data sets, and report to your team's GitHub repository by 11:59 PM on **04/28**.
- Every team member must upload a copy of the report that has all team members' names to Canvas by 11:59 PM on **04/28**.
- Be sure to include the URL to your team's repository on the first page after the title and author's names. Your team's repository is *not* a reference.

Grading

- You will be graded on:
  1. Your ability to choose, apply, and interpret one hypothesis testing procedure to answer the research question.

2. Whether the procedure is appropriate and how you justify its appropriateness.
3. How you answer the research question in both statistical and non-statistical terms. The answer is not important, but how you arrived at the answer is crucial.
4. Readability of your `R` code, including comments, and report.

- The report should not exceed 4 pages, excluding references; otherwise you will be penalized. However, there's no minimum page length (the shorter the better ☺). This page limit is lower than the Part 1 report, from which it should be clear that the Part 2 report is a separate document that should not include your Part 1 report in it.
- There are no bonus point opportunities.
- I will deduct 4 points for every hour (rounded up) that your deliverables are late, either to GitHub or Canvas. 04/28 is the last day of the semester, so I cannot make exceptions without documentary justification.

Part 2 Report

- The report must be a `PDF` file knit directly from the provided project template `.Rmd` file. The report should include the following labeled sections:
  1. Introduction: Reintroduce readers to the chosen topic and research question. Briefly recap the exploratory findings from Part 1 of the project.
  2. Methods:
     (a) You are to pick one research question and one standalone hypothesis testing procedure to answer the research question.
     (b) State the procedure and explain why it is appropriate.
     (c) Verify that all assumptions are met for the test. For a two-sample test, assumptions for each population have to be verified.
     (d) "Large sample size" is not an adequate justification for tests relying on CLT or normality. Expert sources or graphics are appropriate.
     (e) Do not use additional hypotheses tests to verify assumptions.
  3. Results:
     (a) Display all code and output of the test.
     (b) Interpret important components of the test output.
     (c) State whether the test is statistically significant.
  4. Conclusions:
     (a) Interpret the test results in non-statistical terms.
     (b) Explain limitations of the analysis.
     (c) Explain how the conclusions can be generalized beyond the data set.
  5. References:
     (a) This section is not considered part of the 3-page report length limit.
     (b) Your team's repository is *not* a reference!
     (c) Begin this section on a new page.

(d) Include sources used for background research and the data sources.

(e) If you used AI to produce any portion of any file in your repository, you must provide attribution in accordance with UVA guidelines.